

Central Limit Theorem [CLT]

population dist.



Sample	$\bar{X}$ (mean)
$n_1 = 25$	$\bar{X}_1 \quad 4$
$n_2 = 25$	$\bar{X}_2 \quad 3.5$
$n_3 = 25$	$\bar{X}_3 \quad 5.1$
$n_4 = 25$	$\bar{X}_4 \quad 2.8$
$\vdots$	
$n_{30} = 25$	$\bar{X}_{30} \quad 3.9$

$[\bar{X}_1, \bar{X}_2, \bar{X}_3, \dots, \bar{X}_{30}]$

Sample size ≥ 25

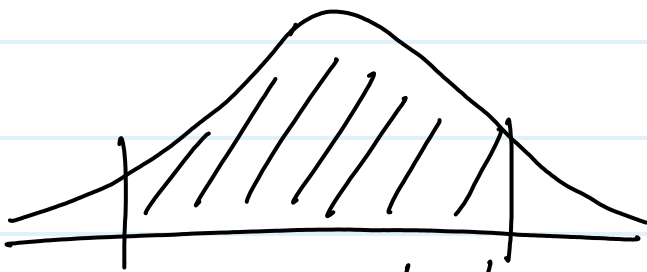
more sample the more normal distribution.



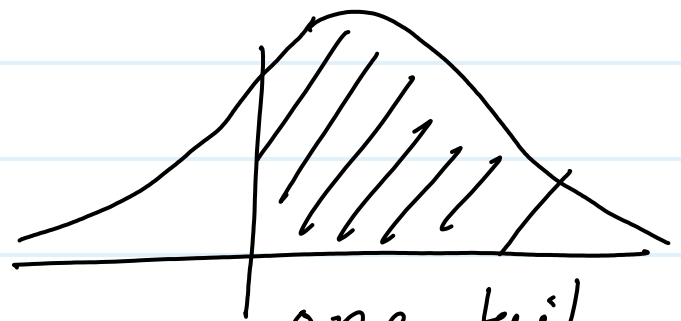
H_0 - Null ✓ →
 H_1 - Alternate ✓ →

$\mu_1 \neq \mu_2$ - Alternate H_1

$\mu_1 = \mu_2$ = null



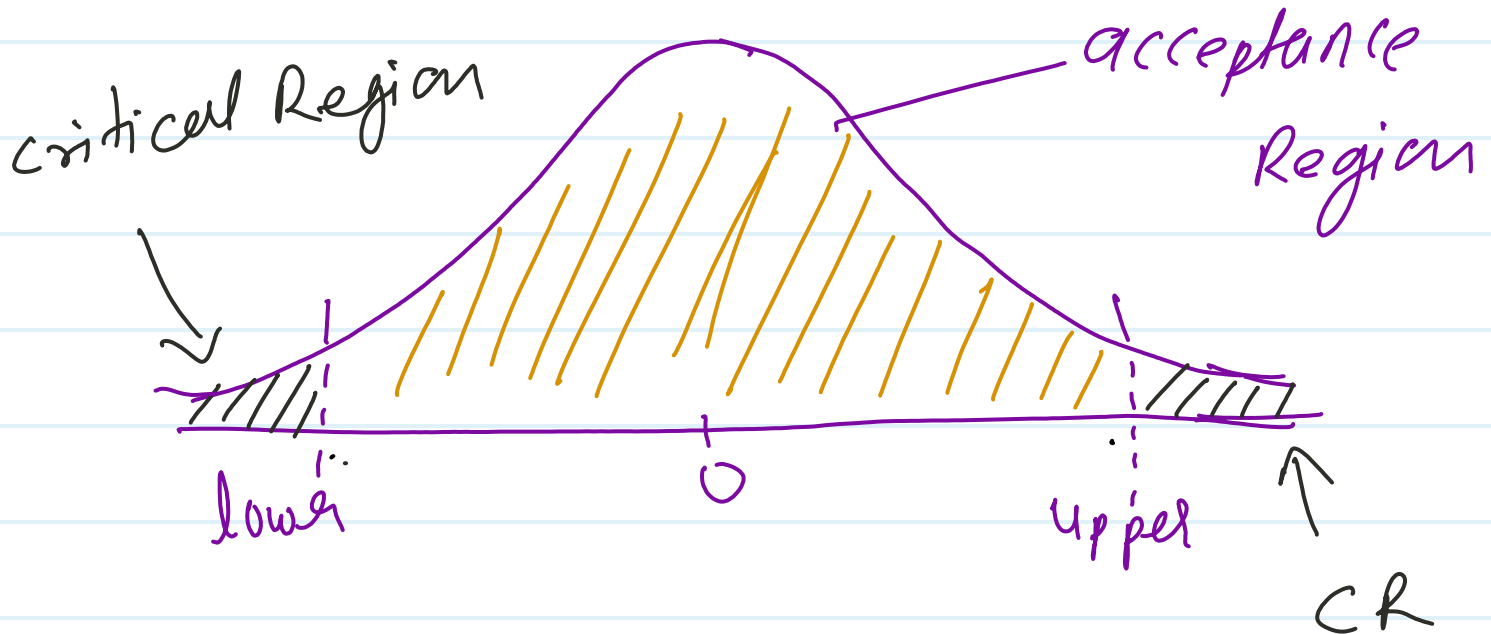
Two tail



one tail

Hypothesis testing

* Confidence interval -



H_0 (null hypothesis) It support always confidence Interval. (Acceptance Region)

H_1 (Alternative hypothesis) It support always critical Region.

✓ H_0 - We fail to reject H_0 .

✓ H_1 = we reject H_0 , and accept H_1 .

$$N = 10000$$

$$\mu = 285$$

$$n = 40$$

$$\bar{x} = 140$$

Two tail -



$$\mu = \bar{x} = \text{Null hypothesis}$$

$$\mu \neq \bar{x} = \text{Alternate hypothesis.}$$

α = Critical Region (significant value)

C.R = Acceptance Region

P = It is probability of H_0 to be True.

$$P = 0.15$$

$$\alpha = 0.05$$

$$P > \alpha$$



In this case we fail to reject null hypothesis.

$$P < \alpha$$

We reject null hypothesis and accept alternate hypothesis.

* Type of hypothesis test

(i) Parametric test

(ii) Non-parametric test

① Parametric test

① Z-test

② T-test

③ Binomial test

④ Poisson test

⑤ Exponential test

② Non-parametric test

① Chi-square test

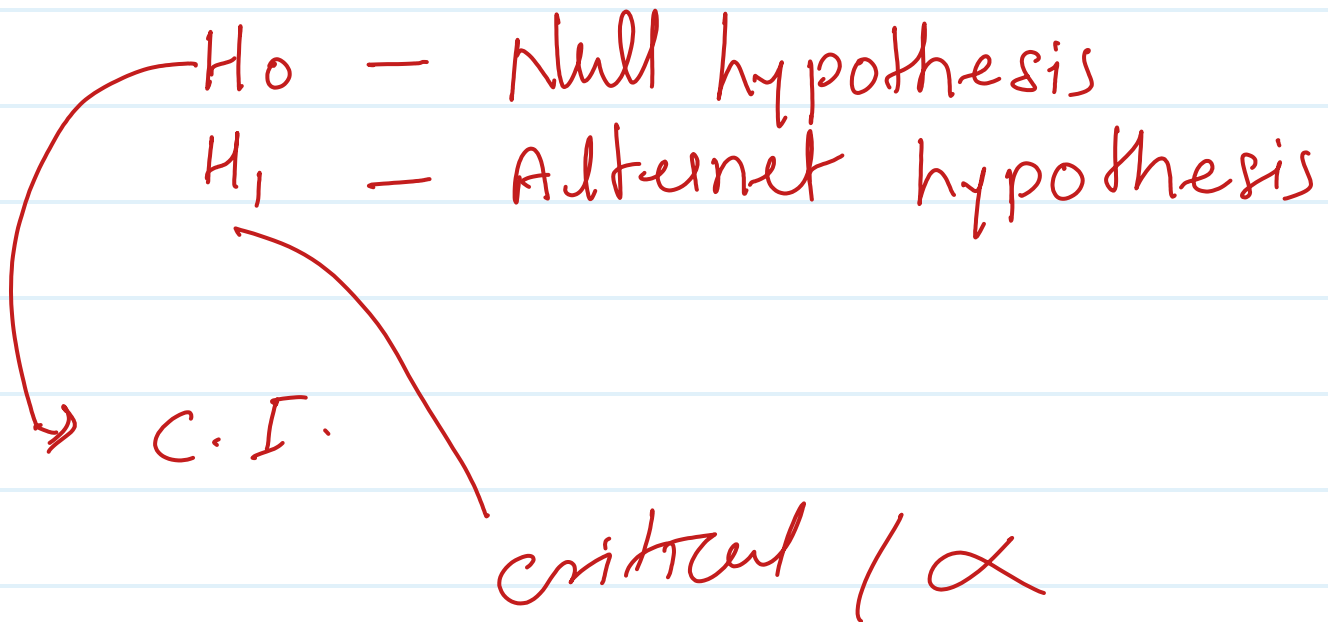
② Anova (F1)

{ Sample size < 30
SD σ missing } t-test

{ Sample size ≥ 30
 σ given } Z-test.



* Hypothesis testing



✓ Doctor claim — 80% effective
 $\Rightarrow \boxed{8/10}$

p value
 to be true of
Alternative hypothesis or null hypothesis

